

#1.a

$$P((A-B) \cup (B-A) \cup (A \cap B)) = P(A-B) + P(B-A) + P(A \cap B)$$

Due to $P(A-B)$, $P(B-A)$, $P(A \cap B)$ are disjoint

$$\Rightarrow P(A \cup B) = P(A-B) + P(B-A) + P(A \cap B) \quad \text{--- ①}$$

$$P(A) + P(B) = P((A-B) \cup (A \cap B)) + P((B-A) \cup (A \cap B))$$

$$= P(A-B) + P(A \cap B) + P(B-A) + P(A \cap B)$$

$$= P(A-B) + P(B-A) + 2P(A \cap B) \quad \text{--- ②}$$

$$\text{②} - \text{①} = P(A) + P(B) - P(A \cap B) = P(A \cup B)$$

$$\Rightarrow P(A) + P(B) = P(A \cap B) + P(A \cup B)$$

#1.b

$P(A \cup B) \geq P(A) + P(B)$ is wrong

From 1.a we can see $P(A) + P(B) = P(A \cup B) + P(A \cap B)$

Due to $P(A \cap B) \geq 0$

so $P(A) + P(B) \geq P(A \cup B)$

Using the inductive assumption, this can be simplified to:

$$\leq \sum_{i=1}^k P(A_i) + P(A_{k+1}) - P\left(\bigcup_{i=1}^k (A_i \cap A_{k+1})\right)$$

Now, by the additivity axiom and the fact that intersections of events are also events:

$$\leq \sum_{i=1}^k P(A_i) + P(A_{k+1}) - \sum_{i=1}^k P(A_i \cap A_{k+1})$$

Due to $-\sum_{i=1}^k P(A_i \cap A_{k+1}) \leq P(\bigcup_{i=1}^k (A_i \cap A_{k+1}))$

So the inequality does not hold.

#2.a

proof.

$$\textcircled{1} \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} S_n \subseteq T$$

pick $x \in \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} S_n$

$\forall k$ from 1 to ∞ there exist n ($n \in [k, \infty)$) st $x \in S_n$

$$\Rightarrow x \in T$$

$$\textcircled{2} T \subseteq \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} S_n \quad \text{pick } x \in T$$

$x \in S_n$ for infinitely many n

then $\forall k \in [1, \infty]$ there must exist n st $x \in S_n$

$$\Rightarrow x \in \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} S_n$$

#2.b

choose $x \in \bigcap_{n=1}^{\infty} S_n \Rightarrow \forall k \quad x$ exist in S_n ($n \geq k$)

Set $H_m \quad m \geq k \Rightarrow x$ exist in S_m

$\Rightarrow$ when m increase, m still $\geq k$ so x still exist in S_m

$\Rightarrow$ and as m increase, it will cover $n \geq k$

$$\Rightarrow \lim_{m \rightarrow \infty} \bigcap_{k=1}^m \bigcup_{n=k}^m S_n = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} S_n$$

#2-C

$$\bigcap_{n=1}^{\infty} A_n = \emptyset \quad \text{pick } x, x \in \mathbb{N}, \forall M \text{ s.t. } x > M$$

we can find a Fibonacci sequence

so we can imagine $\bigcap_{n=1}^{\infty} A_n$ as $(B-C) \cap (C-B)$
and it equals to $\emptyset$ (due to $\{F_k\}_{k \in \mathbb{N}} \neq \mathbb{N}$)

$$\bigcup_{n=1}^{\infty} A_n = (B-C) \cup (C-B)$$

as what mentioned above

$$\bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} A_n = \emptyset \quad \text{whichever } k \text{ we pick}$$

we can find a Fibonacci sequence $> k$

so it equals to $\emptyset$

$$\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n = (B-C) \cup (C-B)$$

as what mentioned above

but it equals to $(B-C) \cup (C-B)$

2.d

① assume we have listed all real numbers

$$x_{i1} = 0. x_{i1} x_{i2} x_{i3} \dots$$

$$0. x_{21} x_{22} x_{23} \dots$$

$$0. x_{i1} x_{i2} x_{i3} \dots$$

② let $y = 0. y_{11} y_{12} y_{13}$

$$y_{ri} = \begin{cases} 2, & \text{if } x_{ri} = 1 \\ 1, & \text{if } x_{ri} \neq 1 \end{cases} \Rightarrow y \text{ would have difference on at least 1 digit with all } x_i$$

$\Rightarrow$ contradict the assumption that we had listed all of real number

3.a

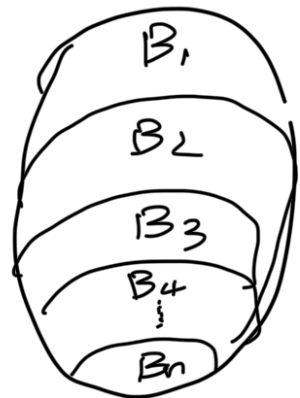
Approach 1

$$B_k = \bigcup_{n=k}^{\infty} A_n$$

$$B_1 \supset B_2 \supset B_3 \supset B_4 \supset \dots \Rightarrow B_k \text{ is decreasing}$$

$$② 0 \leq P\left(\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n\right) = P\left(\bigcap_{k=1}^{\infty} B_k\right)$$

$$= P\left(\lim_{k \rightarrow \infty} B_k\right) = \lim_{k \rightarrow \infty} P(B_k) = \lim_{k \rightarrow \infty} P\left(\bigcup_{n=k}^{\infty} A_n\right) \leq \lim_{k \rightarrow \infty} \sum_{n=k}^{\infty} P(A_n) = 0$$



Approach 2

define $A = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n$

① $P(A_n) \geq P(A_n \cap A) \Rightarrow \sum_n P(A_n) \geq \sum_n P(A_n \cap A)$

② $E(n)$: the number of occurrences of x in A_n

$\lim_{n \rightarrow \infty} E(n) = \infty$

$\sum_{n=1}^{\infty} P(A_n \cap A) = \lim_{n \rightarrow \infty} E(n) \times P(A)$

$\Rightarrow$ if $P(A) > 0$ then $\sum_{n=1}^{\infty} P(A_n) \geq \sum_{n=1}^{\infty} P(A_n \cap A) \geq \lim_{n \rightarrow \infty} E(n) \times P(A) = \infty$ $\neq$

#3.b

① $P(I)$ for $N > 1$

I could be express by $\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n$

$\sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} \frac{1}{2^N} \frac{1}{n^N} < \infty$ (since $N > 1$ by p-series)

$\Rightarrow P(\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n) = 0$ (by Borel-Cantelli Lemma)

② $P(I)$ for $N \leq 1$

$P(\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n) = 1 - P((\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n)^c)$ (let $B_k = \bigcap_{n=k}^{\infty} A_n^c$)

$\Rightarrow \bigcap_{n=k}^{\infty} A_n$ is decreasing, B_k is increasing, $P(\bigcup_{k=1}^{\infty} B_k) = \lim_{k \rightarrow \infty} P(B_k)$

$\Rightarrow 1 - P(\bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} A_n) = 1 - P(\bigcup_{k=1}^{\infty} B_k) = 1 - P(\lim_{k \rightarrow \infty} B_k) = 1 - \lim_{n \rightarrow \infty} P(B_k)$

$$\Rightarrow \lim_{k \rightarrow \infty} P(B_k) \Rightarrow P\left(\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n\right) \text{ (can be any number in } [0, 1])$$

#4

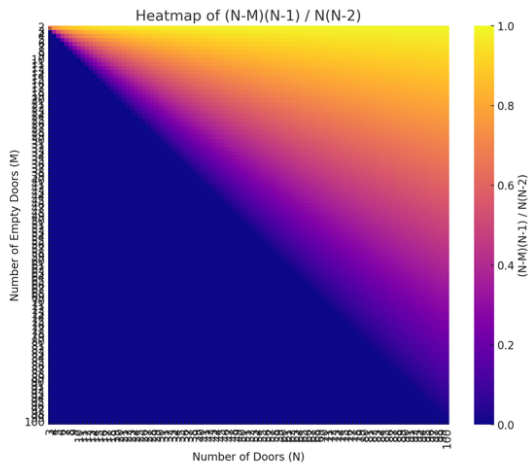
$$\textcircled{1} \quad n \in \{\text{door } 1, \text{door } 2, \text{door } 3, \dots, \text{door } N\}$$

door 1 ~ door N-M mean that these N-M doors are selected to have prize and the rest of doors are selected to empty

$$\textcircled{2} \quad \begin{array}{ll} \text{initial door has prize} & \text{initial door is empty} \\ \Rightarrow \frac{N-M}{N} \times \frac{N-M-1}{N-2} & \Rightarrow \frac{M}{N} \times \frac{N-M}{N-2} \end{array}$$

$$\Rightarrow \frac{(N-M)(N-1)}{N(N-2)}$$

$$\textcircled{3} \quad \begin{array}{ll} \text{當 } n \rightarrow M & P(E_{M,n}) \rightarrow 0, \text{ 因為門都是空} \\ \text{當 } n > M & P(E_{M,n}) \text{ 增加, 空門減少, 有獎勵力} \\ & \text{且門數增加, 故概率率增加} \\ \text{當 } n < M & P(E_{M,n}) \text{ 不存在,} \end{array}$$



#5

$$P(A_i | C) = \frac{P(A_i) P(C | A_i)}{P(C)}$$

$$= \frac{P(A_i) P(C | A_i)}{P(A_1) P(C | A_1) + P(A_2) P(C | A_2) + P(A_3) P(C | A_3)}$$

$$\left[\begin{aligned} P(A_1) P(C | A_1) &= \frac{1}{2} \cdot \left(\frac{1}{4}\right)^6 = \frac{1}{2^{13}} \end{aligned} \right.$$

$$\left[\begin{aligned} P(A_2) P(C | A_2) &= \frac{1}{4} \left(\frac{1}{5} \cdot \frac{1}{2} \cdot \frac{2}{5} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{10} \right) = \frac{1}{2^6} \times \frac{1}{5^3} \end{aligned} \right.$$

$$\left[\begin{aligned} P(A_3) P(C | A_3) &= \frac{1}{4} \left(\frac{1}{5} \cdot \frac{1}{5} \cdot \frac{1}{2} \cdot \frac{1}{5} \cdot \frac{1}{5} \cdot \frac{1}{10} \right) = \frac{1}{5^5} \times \frac{1}{2^4} \end{aligned} \right.$$

$$\Rightarrow P(A_1 | C) = \frac{\frac{1}{2^{13}}}{\frac{1}{2^{13}} + \frac{1}{2^6} \cdot \frac{1}{5^3} + \frac{1}{2^4} \cdot \frac{1}{5^5}} = \frac{3125}{6837}$$

$$P(A_2 | C) = \frac{\frac{1}{2^6} \cdot \frac{1}{5^3}}{\frac{1}{2^{18}} + \frac{1}{2^6} \cdot \frac{1}{5^3} + \frac{1}{2^4} \cdot \frac{1}{5^5}} = \frac{3200}{6837} \text{ \#}$$

$$P(A_3 | C) = \frac{512}{6837} \text{ \#}$$

$\Rightarrow$ the most probable value at Q is $(0.5, 0.2, 0.2, 0.1)$

